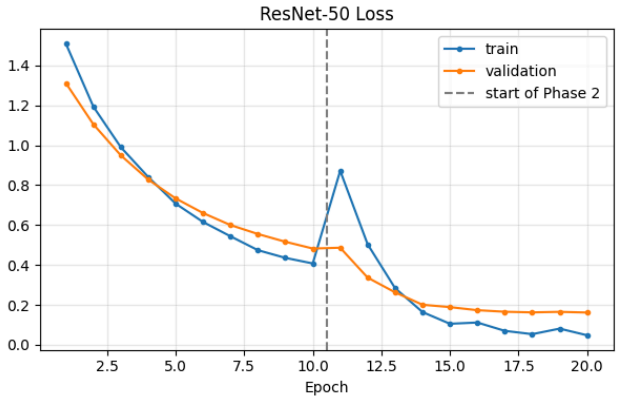
Results:

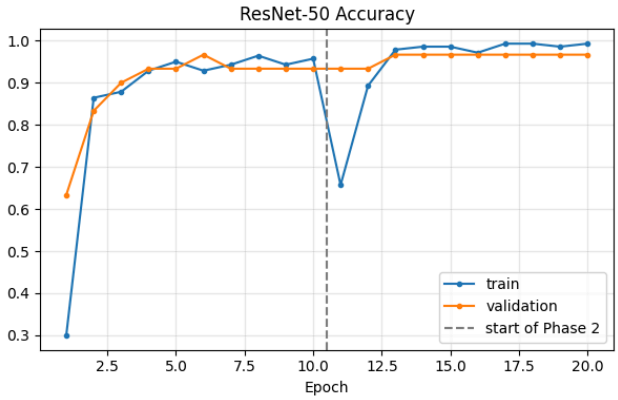
Resnet_results:

```
...  Train Accuracy: 1.0
     Validation Accuracy: 0.9666666666666667
     Test Accuracy: 1.0
     Macro Precision: 1.0
     Macro Recall: 1.0
     Macro F1: 1.0
                    precision    recall  f1-score   support

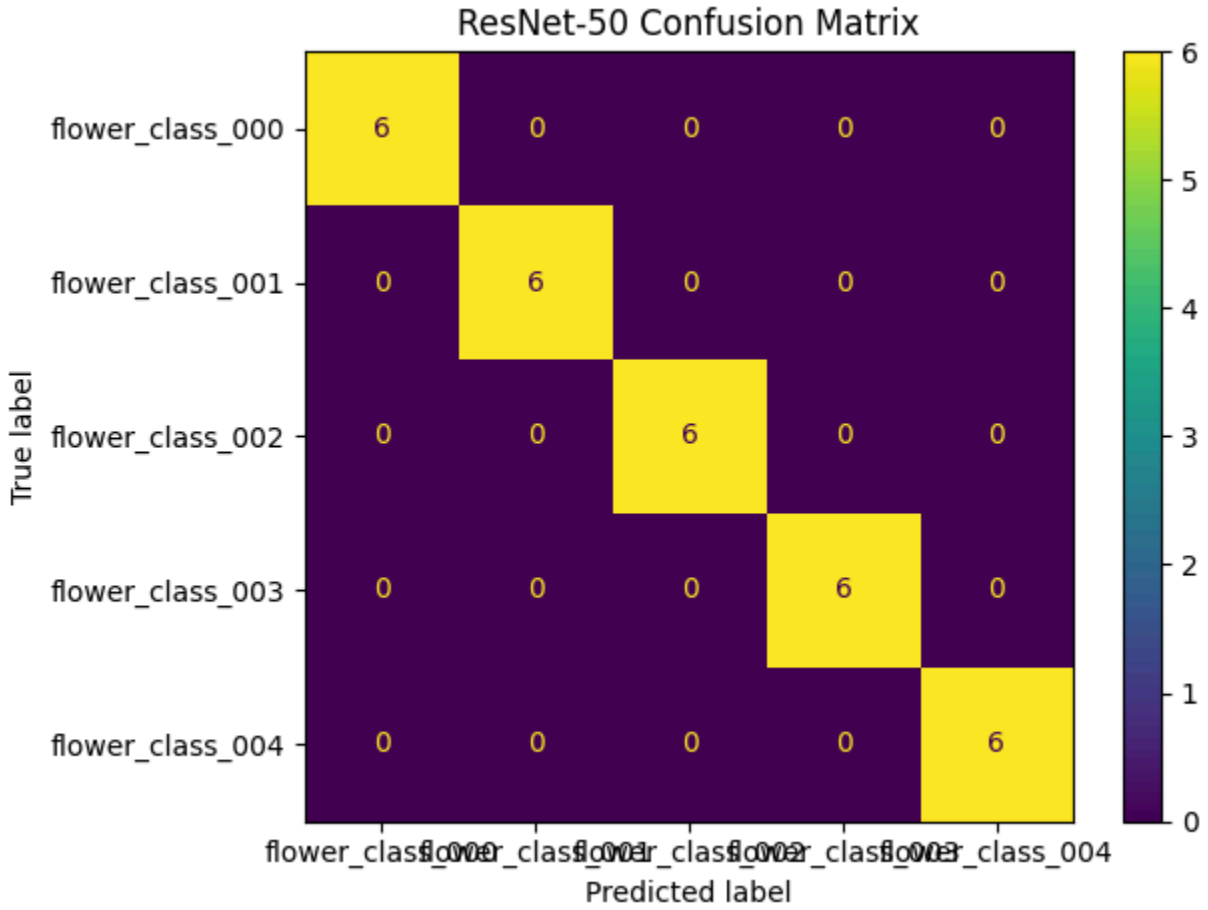
     flower_class_000       1.00      1.00      1.00         6
     flower_class_001       1.00      1.00      1.00         6
     flower_class_002       1.00      1.00      1.00         6
     flower_class_003       1.00      1.00      1.00         6
     flower_class_004       1.00      1.00      1.00         6

            accuracy                           1.00        30
           macro avg       1.00      1.00      1.00        30
        weighted avg       1.00      1.00      1.00        30
```





Resnet_confusion_matrix:

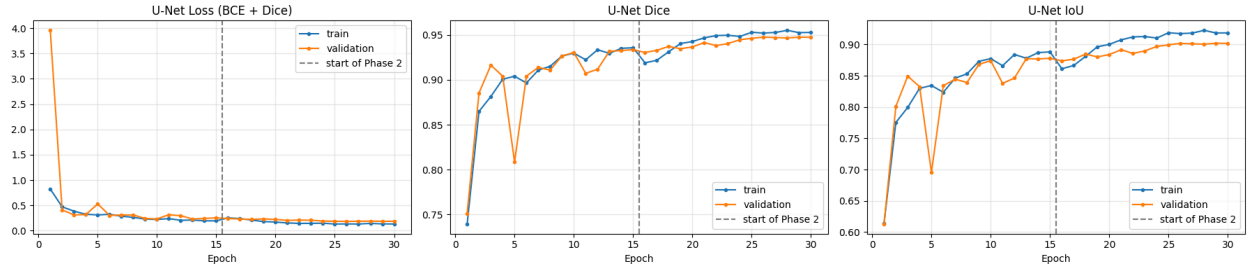



```
Phase 2: trainable parameters: 14,974,981 of 23,518,277
Phase 2: trainable parameters: 14,974,981 of 23,518,277
Epoch 01/10 | train loss 0.8719 acc 0.657 | val loss 0.4863 acc 0.933 | 2.3s
Epoch 02/10 | train loss 0.5036 acc 0.893 | val loss 0.3365 acc 0.933 | 2.2s
Epoch 03/10 | train loss 0.2838 acc 0.979 | val loss 0.2627 acc 0.967 | 2.4s
Epoch 04/10 | train loss 0.1638 acc 0.986 | val loss 0.2005 acc 0.967 | 3.4s
Epoch 05/10 | train loss 0.1051 acc 0.986 | val loss 0.1886 acc 0.967 | 2.2s
Epoch 06/10 | train loss 0.1113 acc 0.971 | val loss 0.1732 acc 0.967 | 2.1s
Epoch 07/10 | train loss 0.0697 acc 0.993 | val loss 0.1654 acc 0.967 | 2.1s
Epoch 08/10 | train loss 0.0531 acc 0.993 | val loss 0.1622 acc 0.967 | 2.1s
Epoch 09/10 | train loss 0.0808 acc 0.986 | val loss 0.1649 acc 0.967 | 2.2s
Epoch 10/10 | train loss 0.0475 acc 0.993 | val loss 0.1614 acc 0.967 | 4.9s
Best validation accuracy after Phase 2: 0.967
```

Unet_predictions:



Unet_curves:




```
Training loss: 0.1196
Validation   : loss 0.1831 | Dice 0.9477 | IoU 0.9019
TEST         : loss 0.2369 | Dice 0.9315 | IoU 0.8773 | pixel acc 0.9460
```

```
Phase 2 -> trainable parameters: 31,076,177 of 32,521,105
Epoch 01/15 | train loss 0.2532 dice 0.919 | val loss 0.2400 dice 0.931 iou 0.874 | 4.2s
Epoch 02/15 | train loss 0.2392 dice 0.922 | val loss 0.2307 dice 0.933 iou 0.877 | 4.3s
Epoch 03/15 | train loss 0.2089 dice 0.931 | val loss 0.2234 dice 0.937 iou 0.885 | 4.1s
Epoch 04/15 | train loss 0.1802 dice 0.941 | val loss 0.2336 dice 0.935 iou 0.880 | 4.2s
Epoch 05/15 | train loss 0.1705 dice 0.943 | val loss 0.2211 dice 0.937 iou 0.884 | 5.0s
Epoch 06/15 | train loss 0.1536 dice 0.947 | val loss 0.2044 dice 0.942 iou 0.892 | 4.2s
Epoch 07/15 | train loss 0.1431 dice 0.949 | val loss 0.2100 dice 0.938 iou 0.886 | 3.7s
Epoch 08/15 | train loss 0.1449 dice 0.950 | val loss 0.2062 dice 0.940 iou 0.890 | 3.7s
Epoch 09/15 | train loss 0.1478 dice 0.948 | val loss 0.1912 dice 0.945 iou 0.897 | 6.4s
Epoch 10/15 | train loss 0.1316 dice 0.953 | val loss 0.1860 dice 0.946 iou 0.900 | 3.7s
Epoch 11/15 | train loss 0.1335 dice 0.952 | val loss 0.1831 dice 0.948 iou 0.902 | 6.1s
Epoch 12/15 | train loss 0.1319 dice 0.953 | val loss 0.1863 dice 0.947 iou 0.901 | 4.3s
Epoch 13/15 | train loss 0.1439 dice 0.955 | val loss 0.1893 dice 0.947 iou 0.901 | 3.5s
Epoch 14/15 | train loss 0.1342 dice 0.953 | val loss 0.1861 dice 0.948 iou 0.902 | 7.4s
Epoch 15/15 | train loss 0.1327 dice 0.953 | val loss 0.1862 dice 0.947 iou 0.902 | 5.9s
Best validation Dice after Phase 2: 0.948
```

Csv:

Results Tables (fill in with YOUR results)

ResNet-50 Metric	Result
Training Accuracy	**1.0**
Validation Accuracy	*0.967__*
Test Accuracy	**1.0**
Precision (macro)	**1.0**
Recall (macro)	*1.0_*
F1-score (macro)	*1.0_*

U-Net Metric	Result
Training Loss	*0.1196*
Validation Loss	*0.1831_*
Test Dice	_0.9315 _
Test IoU	*0.8773*

Model	Dataset	Task	Main Metric	Result
ResNet-50	*oxford flowers_*	Classification	Accuracy	*1.0_*
U-Net	*oxford iit pet*	Segmentation	Dice / IoU	*0.93/0/87__*